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Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2022

Time: 03 Hours

Code: 105504 ***Subject: Software Engineering*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven questions only): **[14]**

(a) The spiral model was originally proposed by
- (i) IBM
- (ii) Barry Boehm
- (iii) Pressman
- (iv) Royce

(b) Measure of reliability is given by
- (i) mean time between success
- (ii) MTBF
- (iii) mean reliable
- (iv) MTTR

(c) Which of the following is not a use of a CASE tool?
- (i) It supports structured analysis and design (SA/SD)
- (ii) It maintains the data dictionary
- (iii) It checks whether DFDs are balanced or not
- (iv) It compiles with the available system

(d) Name of an evaluation technique to assess the quality of test cases is
- (i) mutation analysis
- (ii) validation
- (iii) verification
- (iv) performance analysis

(e) What is the most popular model for student program?
- (i) Waterfall model
- (ii) Built-and-fix model
- (iii) Spiral model
- (iv) Rational unified model

(f) Which of the following is not a part of bug report?
- (i) Test case
- (ii) Output
- (iii) Software version
- (iv) LOC

(g) Independent modules are easier to maintain and test because of
- (i) code modification is limited
- (ii) reusable modules are possible
- (iii) error propagation is reduced
- (iv) All of the above

(h) In size-oriented metrics, metrics are developed based on the
- (i) number of functions
- (ii) number of user inputs
- (iii) number of lines of code
- (iv) amount of memory usage

(i) Classes communicate with one another via
- (i) processed information
- (ii) interfaces
- (iii) messages
- (iv) coupling

(j) Software is not considered to be collection of executable programming code, associated libraries and documentations.
- (i) Statement is true
- (ii) Software is only data structures with algorithms
- (iii) Statement is false
- (iv) Statement underestimates software

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Q.2 Solve this question : **[14]**

Explain in detail the classical waterfall model with help of a neat and labelled diagram.

Q.3 Solve both questions : **[7+7=14]**

(a) What is prototyping model? Explain the problems and advantages of prototyping in detail.

(b) Develop a test case for any testing technique for 'student admission system'.

Q.4 Solve both questions : **[7+7=14]**

(a) Define cohesion and coupling. Explain various types of each of them.

(b) What are CASE tools? With a suitable diagram, explain the categories of CASE tools.

Q.5 Solve this question : **[14]**

What are the different COCOMO models? Explain the phases involved in the detailed COCOMO model.

Q.6 Solve both questions : **[7+7=14]**

(a) What are the types of user-interface design?

(b) Explain the stages of object-oriented design process.

Q.7 Write short notes on the following: **[14]**

(a) Unified modelling language

(b) Object-oriented analysis modelling

(c) Object-oriented design concepts and methods

Q.8 Solve this question : **[14]**

Explain coding standards, coding guidelines and code review techniques in detail.

Q.9 Solve both questions : **[7+7=14]**

(a) What is software quality? Discuss software quality attributes.

(b) Discuss the differences between object-oriented and function-oriented designs.

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